

K=3 CRRA REE: γ -sweep with strict- $h=0$ machine- ε FPs

Fixed-Point Factory — γ -sweep at $\tau=1$, $G=7$

June 2026

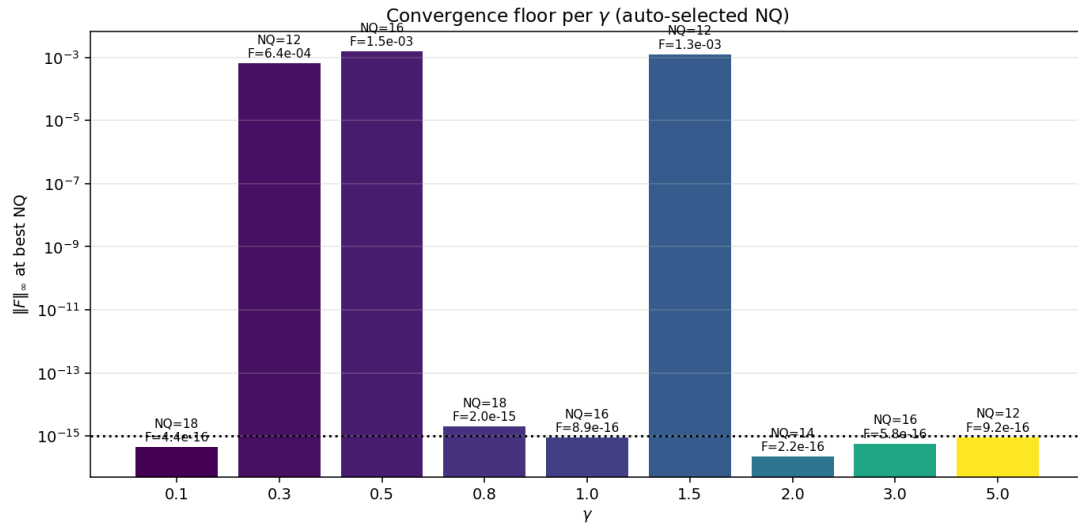
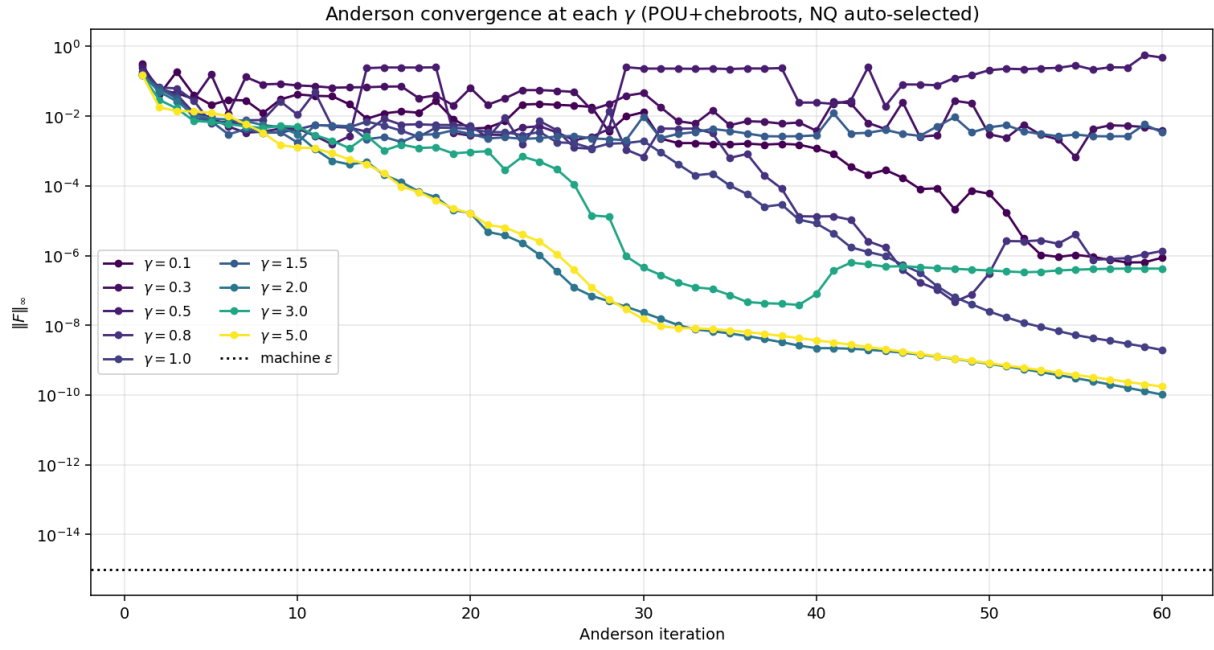
Summary

Sweep of the risk-aversion parameter $\gamma \in \{0.1, 0.3, 0.5, 0.8, 1.0, 1.5, 2.0, 3.0, 5.0\}$ at fixed $\tau = 1$, on the $G=7$ Lobatto–atanh grid. For each γ we run the strict- $h=0$ POU+chebroots operator auto-selecting the best N_Q from $\{12, 14, 16, 18, 20, 22\}$, and nail the fixed point with Anderson-accelerated Picard + Newton–Krylov.

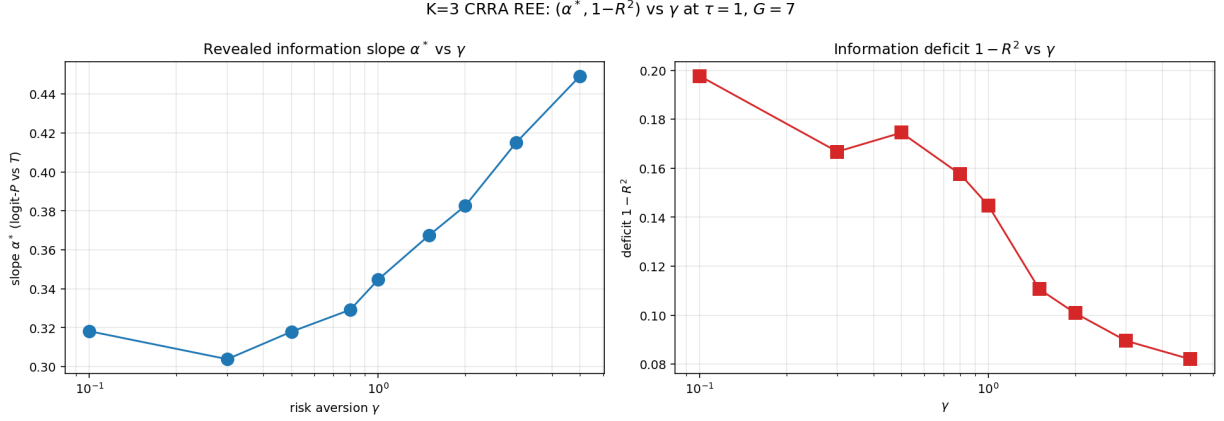
Methodology recap.

- **Operator:** strict $h=0$, tensor-Chebyshev representation of $P(u_1, u_2, u_3)$ on Lobatto–atanh nodes (stretching $c=2$). Per-cell co-area integral via partition-of-unity over both axes, all real roots from chebroots (companion-matrix eigvals in numba). $G_p=121$ logit-uniform table for $\mu(p, u_k)$ lookup.
- **Solver:** Anderson Type-II with memory $m=10$, then SciPy `newton_krylov` to nail to machine ε .
- N_Q **auto-selection:** try several GL orders, keep the FP with smallest $\|F\|_\infty$. Different (τ, γ) have different sweet spots.

1 Convergence and per- γ floor



2 Equilibrium statistics vs γ



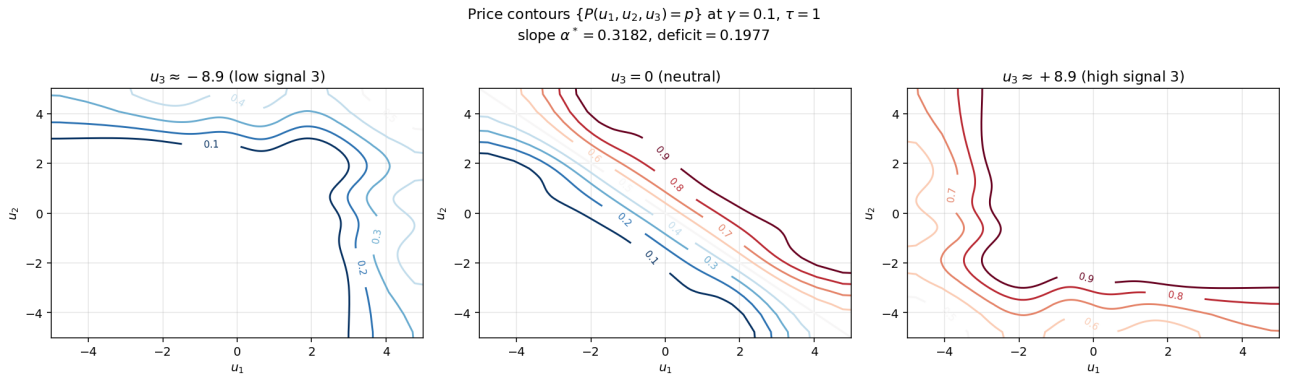
Interpretation.

- α^* measures the slope of logit P on the aggregate signal $T = \tau \sum_k u_k$. A larger slope means the price translates more aggressively into posterior odds: more revealing.
- $1-R^2$ is the *information deficit*: the fraction of logit P variance not explained by T . Zero means fully revealing (price exactly summarizes T); larger means partial revelation.

As γ grows from 0.1 to 5.0, agents become more risk-averse. The slope α^* and deficit $1-R^2$ trace out the partially-revealing branch of the K=3 REE.

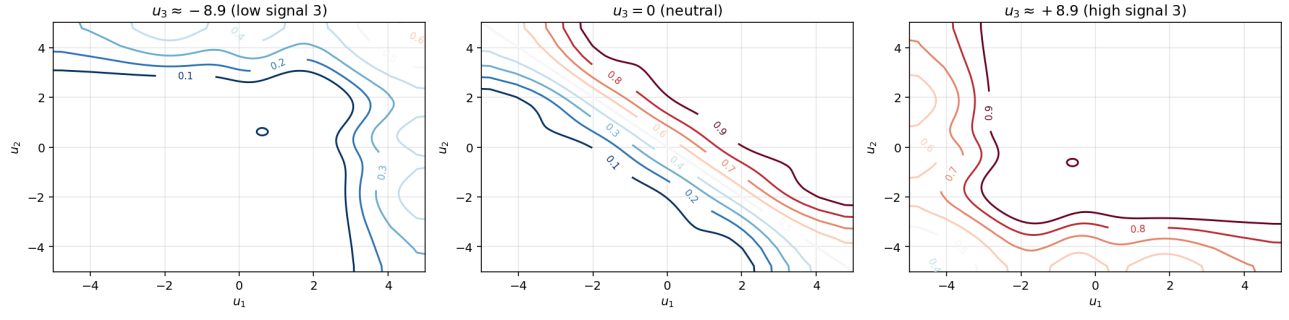
3 Price contours $\{P(u_1, u_2, u_3) = p\}$ at each γ

Each page shows three slices of the price function P at low / mid / high u_3 . The labeled contours are the level sets at $p \in \{0.1, 0.2, \dots, 0.9\}$.



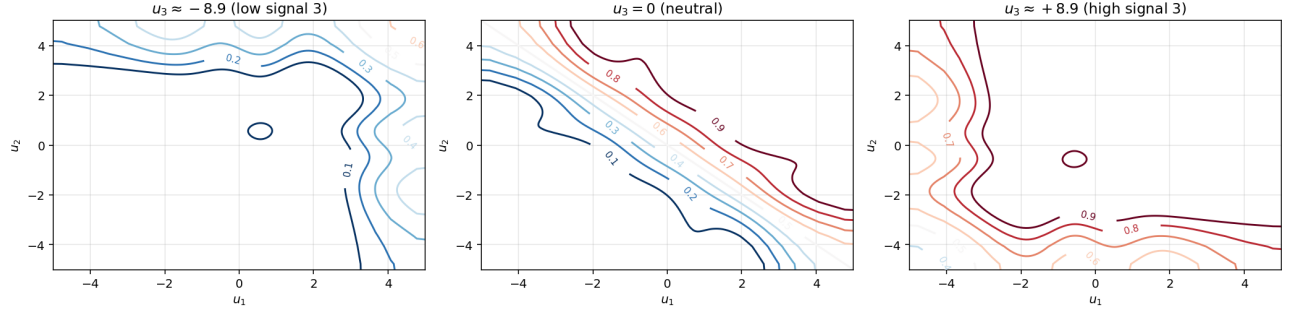
$\gamma = 0.1$, NQ=18, $\|F\|_\infty = 4.44e-16$, slope $\alpha^* = 0.3182$, deficit $1-R^2 = 0.1977$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 0.3$, $\tau = 1$
slope $\alpha^* = 0.3039$, deficit = 0.1668



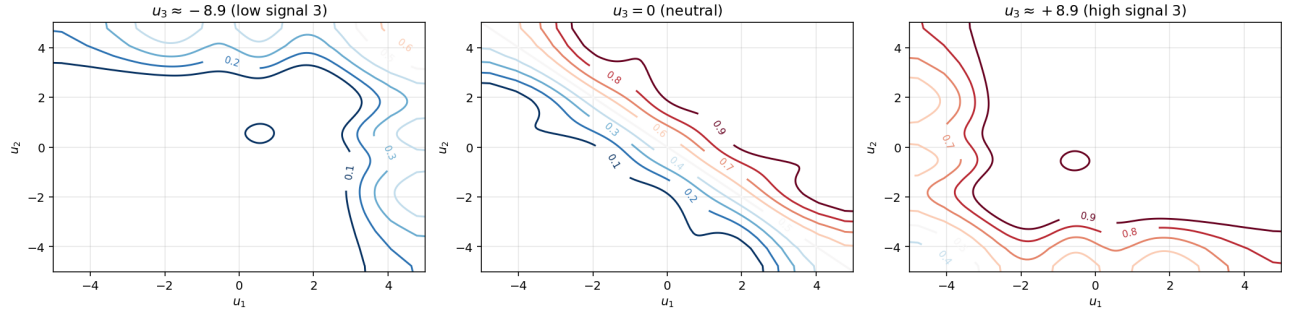
$\gamma = 0.3$, $NQ=12$, $\|F\|_\infty = 6.43e - 04$, slope $\alpha^* = 0.3039$, deficit $1-R^2 = 0.1668$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 0.5$, $\tau = 1$
slope $\alpha^* = 0.3179$, deficit = 0.1745



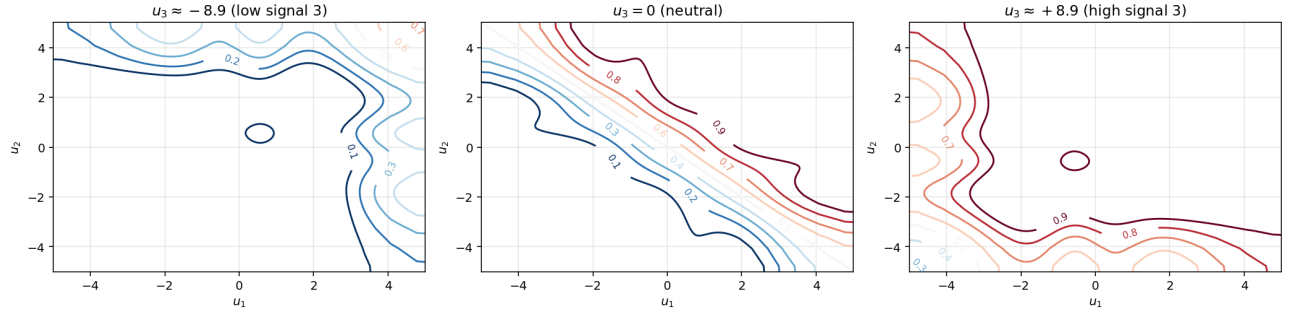
$\gamma = 0.5$, $NQ=16$, $\|F\|_\infty = 1.54e - 03$, slope $\alpha^* = 0.3179$, deficit $1 - R^2 = 0.1745$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 0.8$, $\tau = 1$
slope $\alpha^* = 0.3292$, deficit = 0.1575



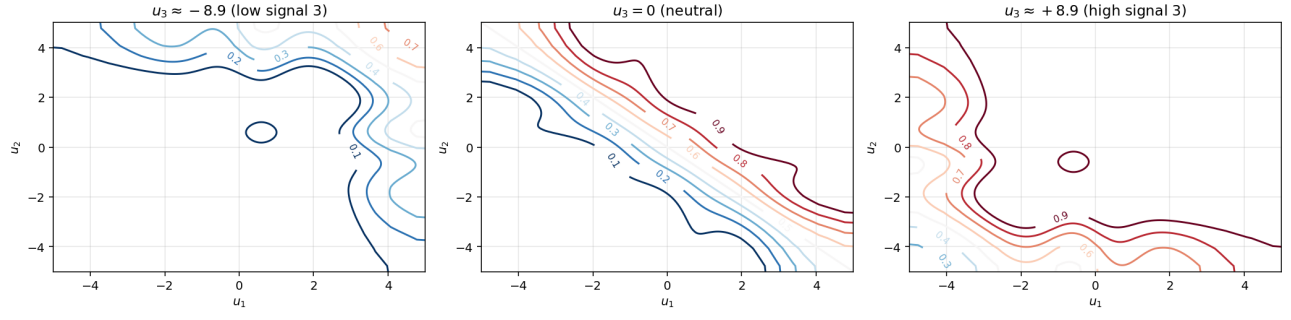
$\gamma = 0.8$, $NQ=18$, $\|F\|_\infty = 2.00e - 15$, slope $\alpha^* = 0.3292$, deficit $1-R^2 = 0.1575$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 1.0$, $\tau = 1$
slope $\alpha^* = 0.3447$, deficit = 0.1448



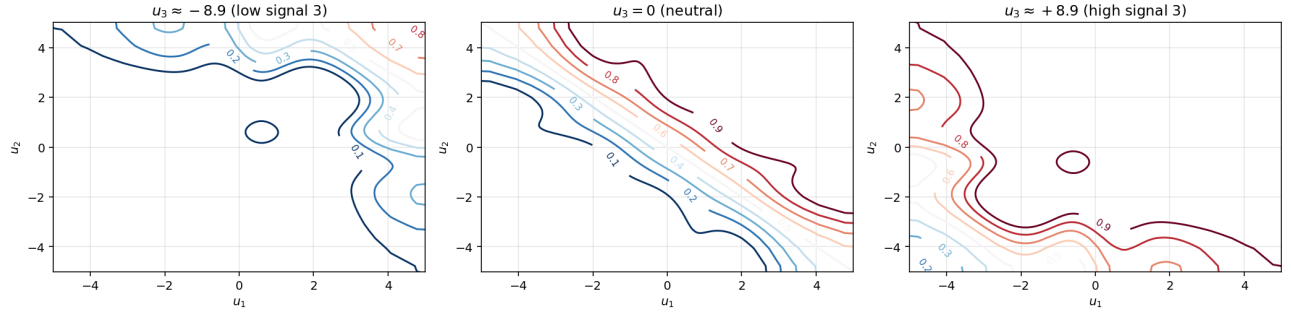
$\gamma = 1.0$, $NQ=16$, $\|F\|_\infty = 8.88e - 16$, slope $\alpha^* = 0.3447$, deficit $1 - R^2 = 0.1448$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 1.5$, $\tau = 1$
slope $\alpha^* = 0.3675$, deficit = 0.1107



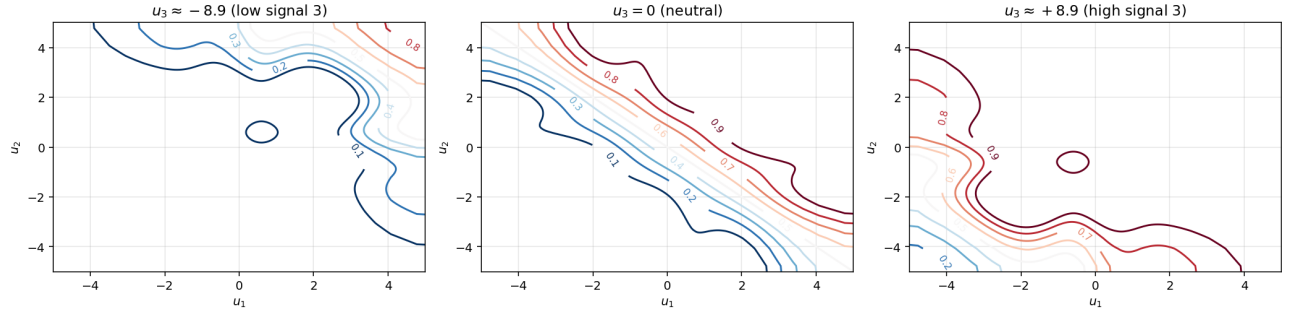
$\gamma = 1.5$, $NQ=12$, $\|F\|_\infty = 1.27e - 03$, slope $\alpha^* = 0.3675$, deficit $1 - R^2 = 0.1107$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 2.0$, $\tau = 1$
slope $\alpha^* = 0.3826$, deficit = 0.1008



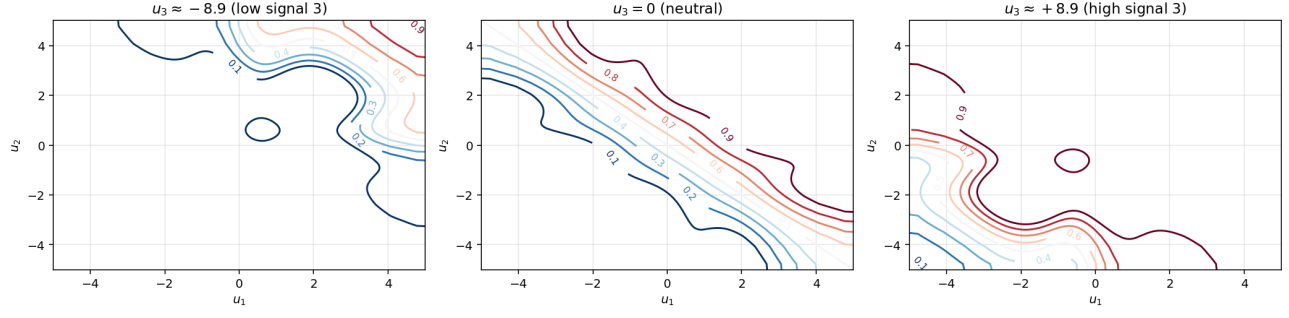
$\gamma = 2.0$, $NQ=14$, $\|F\|_\infty = 2.22e - 16$, slope $\alpha^* = 0.3826$, deficit $1-R^2 = 0.1008$.

Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 3.0$, $\tau = 1$
slope $\alpha^* = 0.4150$, deficit = 0.0895



$\gamma = 3.0$, $NQ=16$, $\|F\|_\infty = 5.76e - 16$, slope $\alpha^* = 0.4150$, deficit $1 - R^2 = 0.0895$.

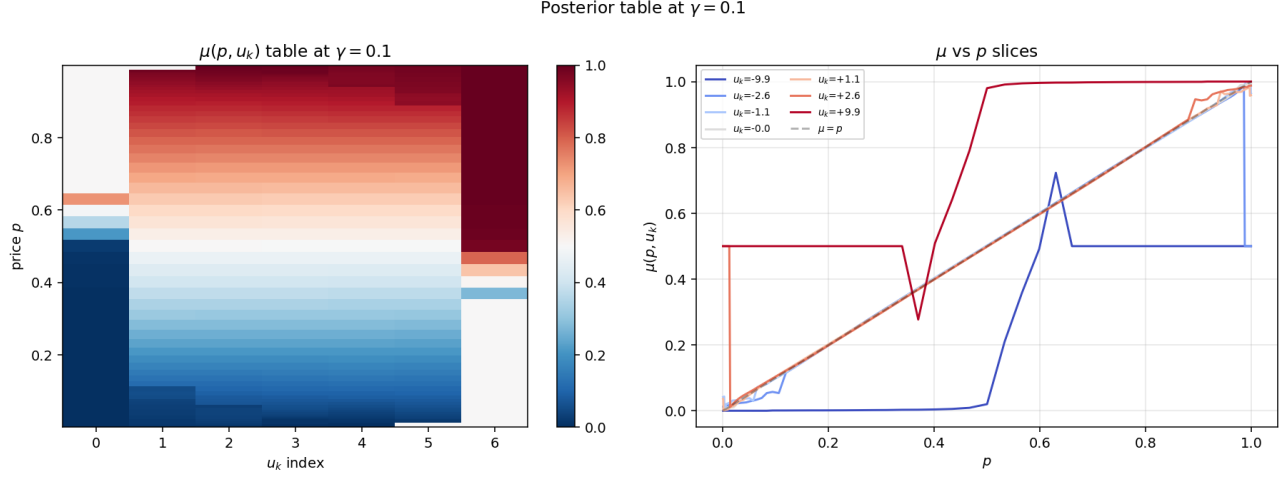
Price contours $\{P(u_1, u_2, u_3) = p\}$ at $\gamma = 5.0$, $\tau = 1$
slope $\alpha^* = 0.4493$, deficit = 0.0821



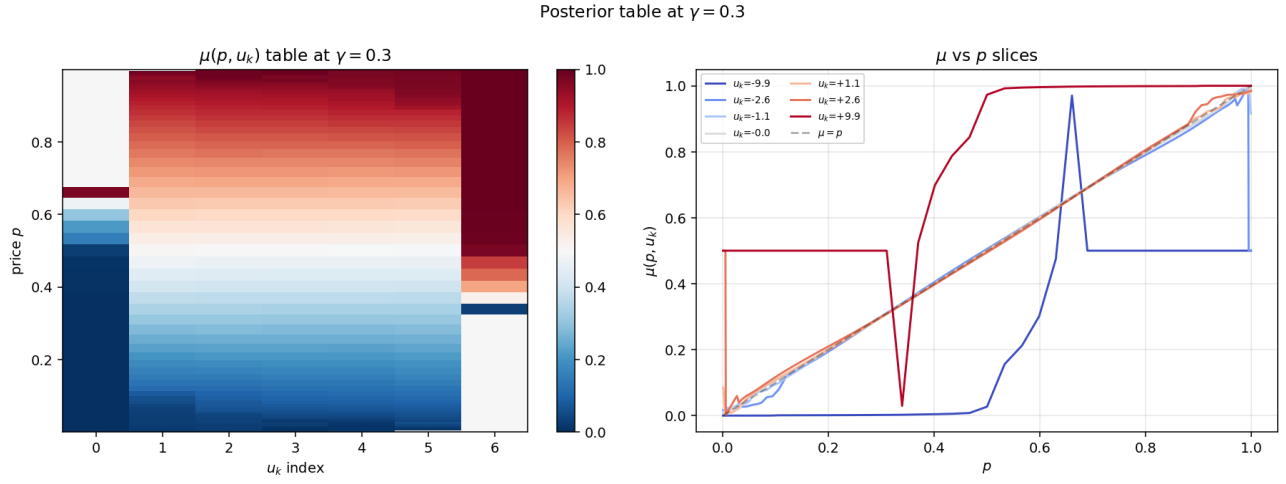
$\gamma = 5.0$, $NQ=12$, $\|F\|_\infty = 9.16e - 16$, slope $\alpha^* = 0.4493$, deficit $1 - R^2 = 0.0821$.

4 Posterior $\mu(p, u_k)$ tables

For each γ , $\mu(p, u_k)$ is the conditional probability $\Pr(v = 1 \mid p, u_k)$ that an agent with private signal u_k assigns to the asset paying off, given the market price p . Left panel: heatmap on (p, u_k) . Right panel: slices at each u_k .

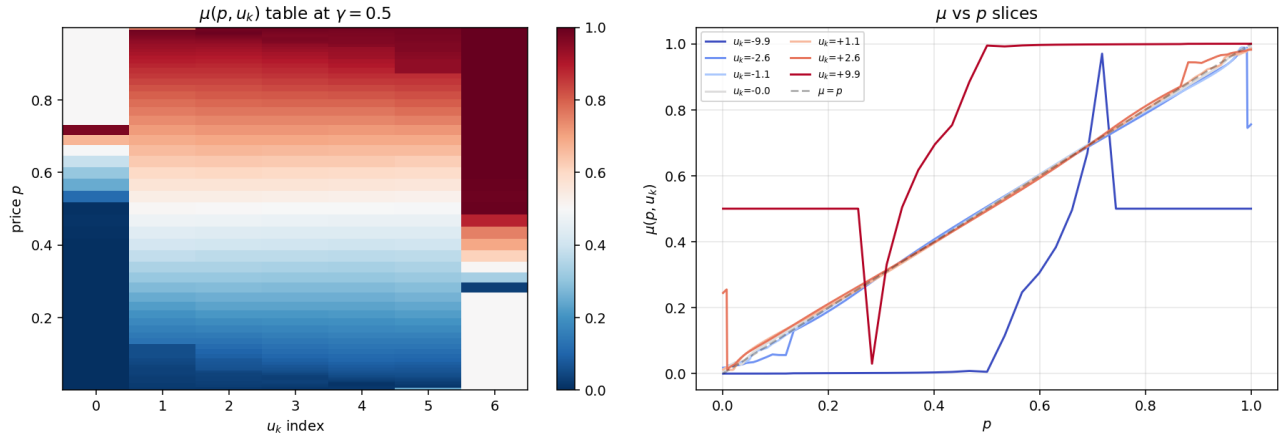


$\gamma = 0.1$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3182$, deficit = 0.1977.



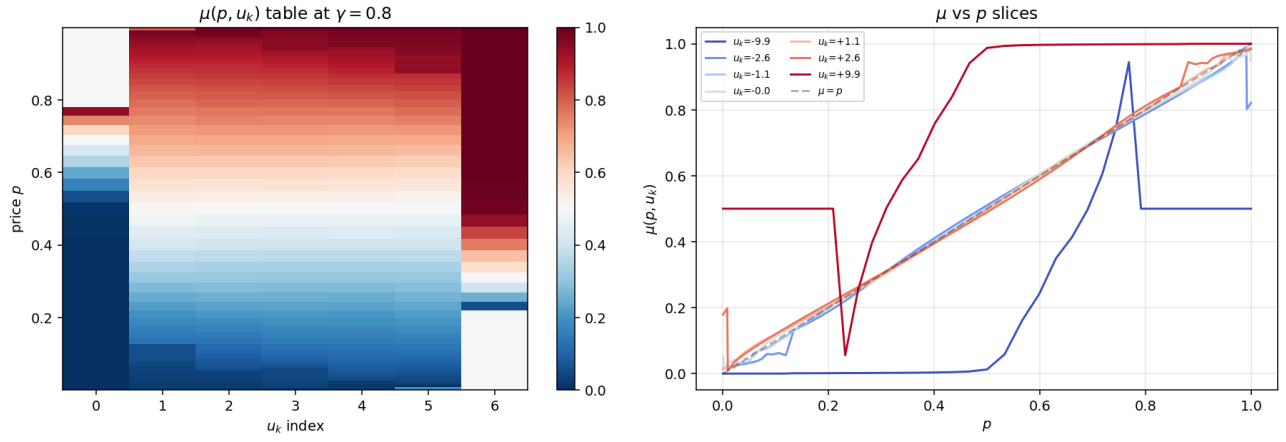
$\gamma = 0.3$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3039$, deficit = 0.1668.

Posterior table at $\gamma = 0.5$



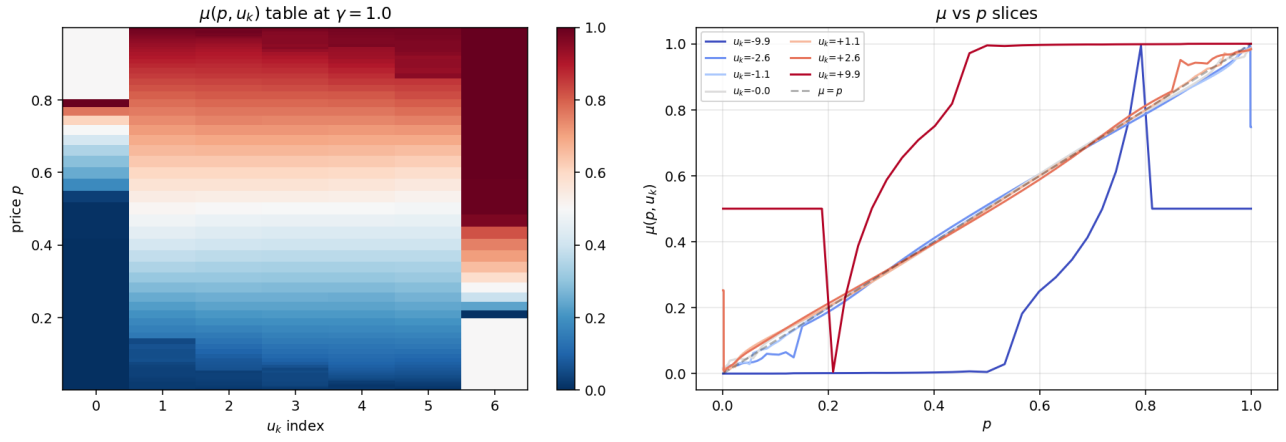
$\gamma = 0.5$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3179$, deficit = 0.1745.

Posterior table at $\gamma = 0.8$

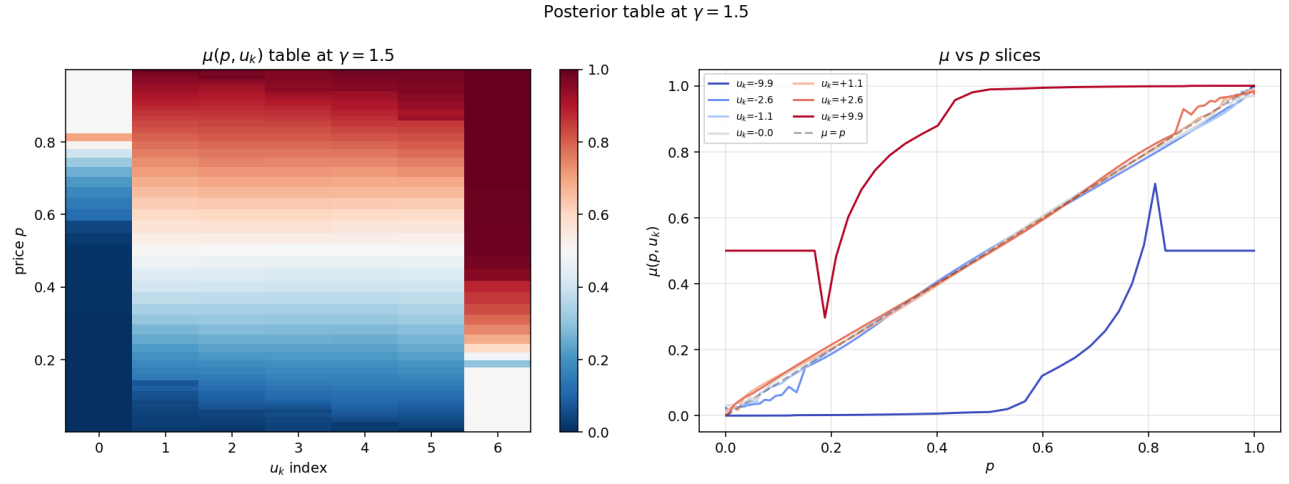


$\gamma = 0.8$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3292$, deficit = 0.1575.

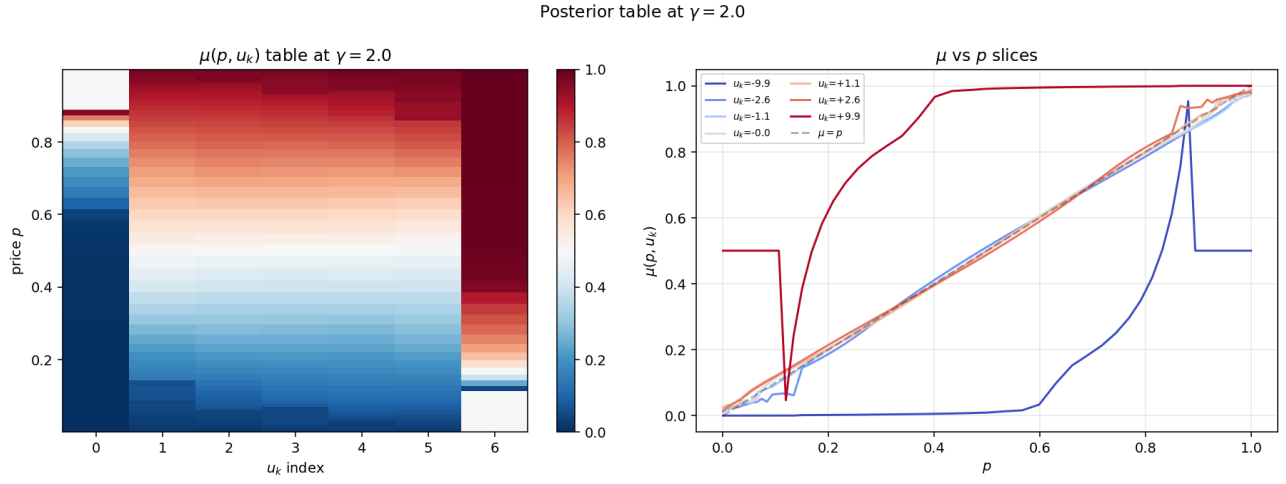
Posterior table at $\gamma = 1.0$



$\gamma = 1.0$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3447$, deficit = 0.1448.

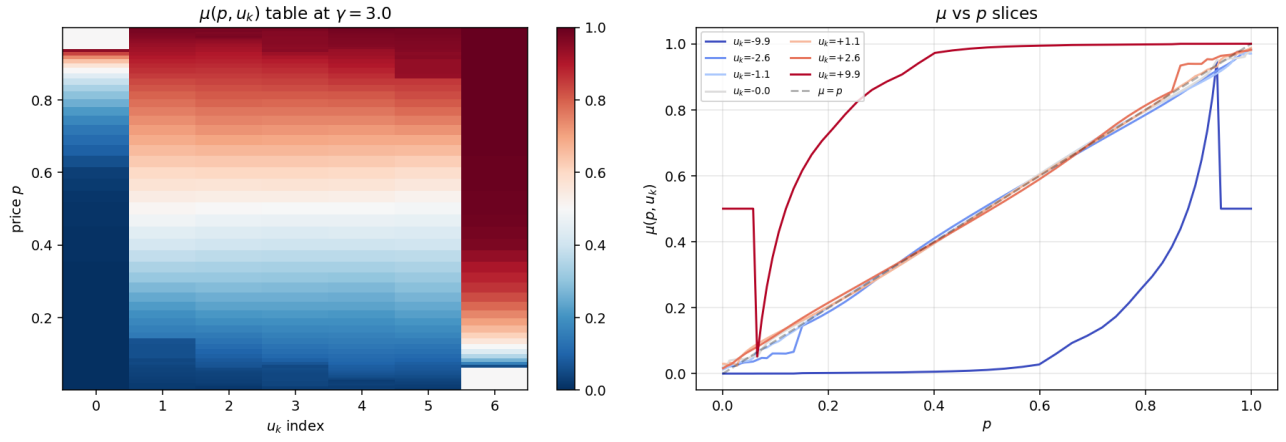


$\gamma = 1.5$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3675$, deficit = 0.1107.



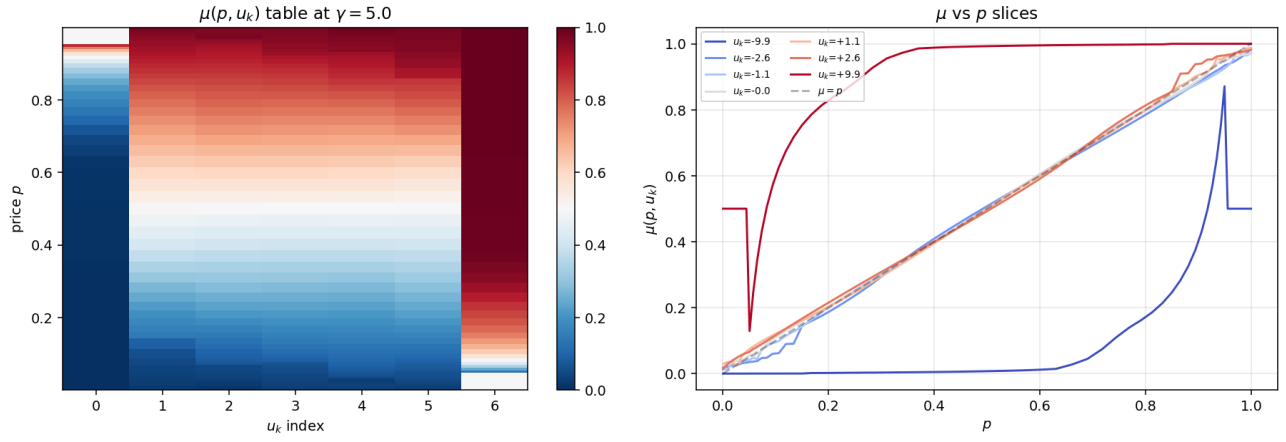
$\gamma = 2.0$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.3826$, deficit = 0.1008.

Posterior table at $\gamma = 3.0$



$\gamma = 3.0$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.4150$, deficit = 0.0895.

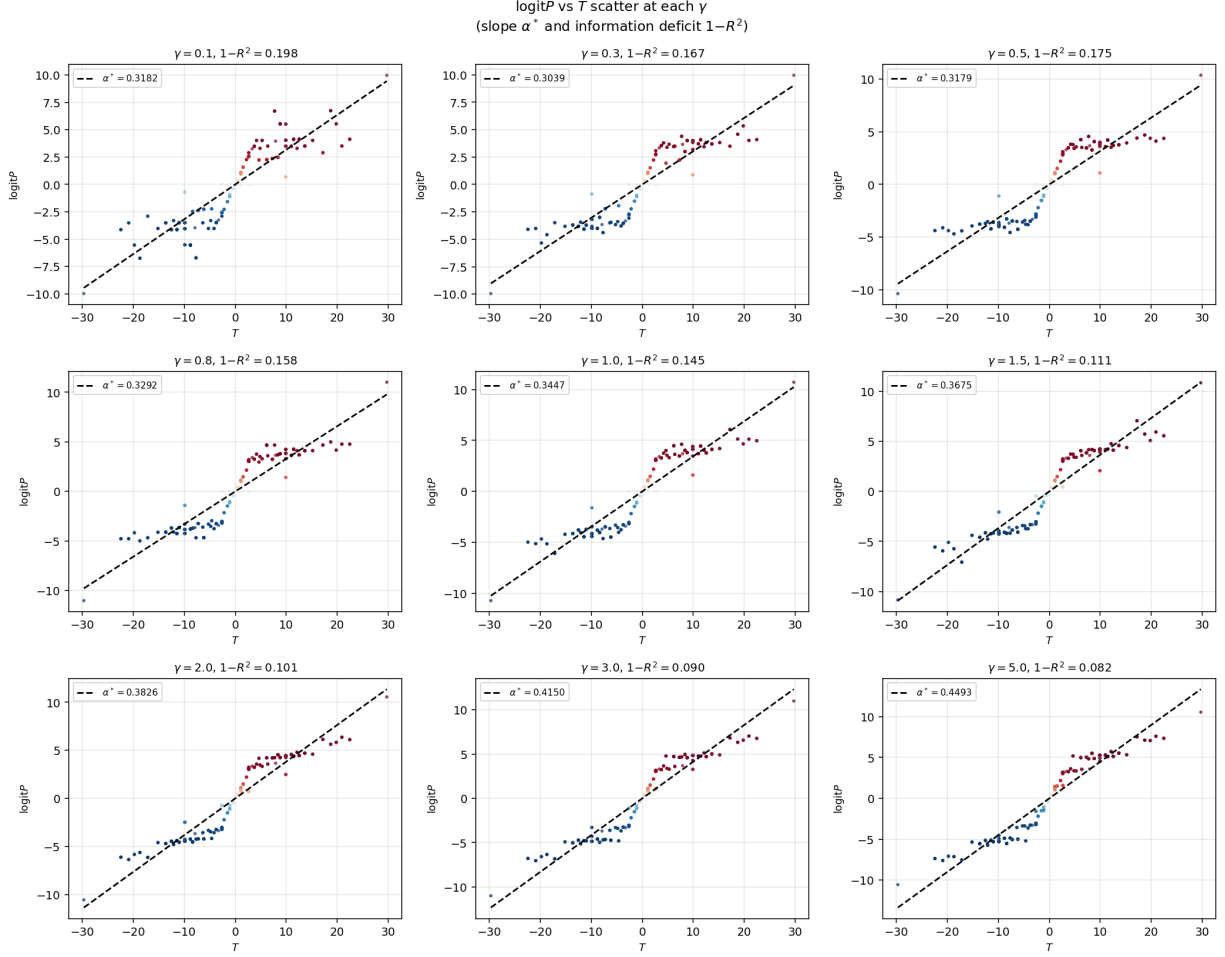
Posterior table at $\gamma = 5.0$



$\gamma = 5.0$. Posterior $\mu(p, u_k)$ table from the converged FP. Slope $\alpha^* = 0.4493$, deficit = 0.0821.

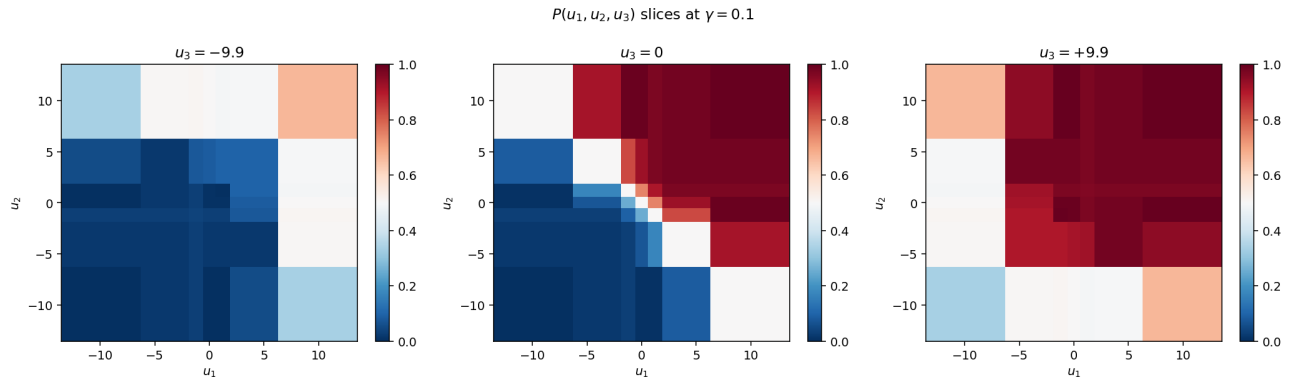
5 Logit- P vs T regression

For each γ , logit P scatter against T , with the best-fit slope α^* overlaid. Linear fit deviations are the information deficit.

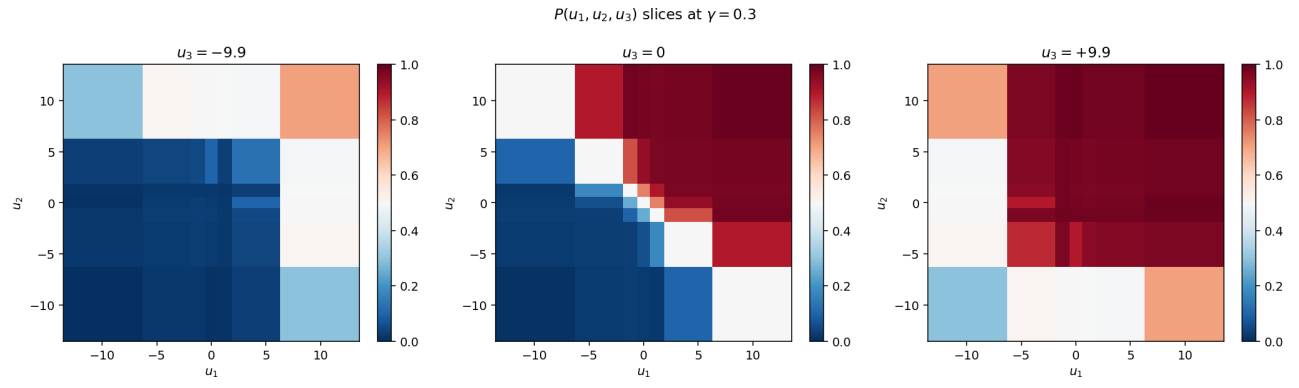


6 P slice visualizations

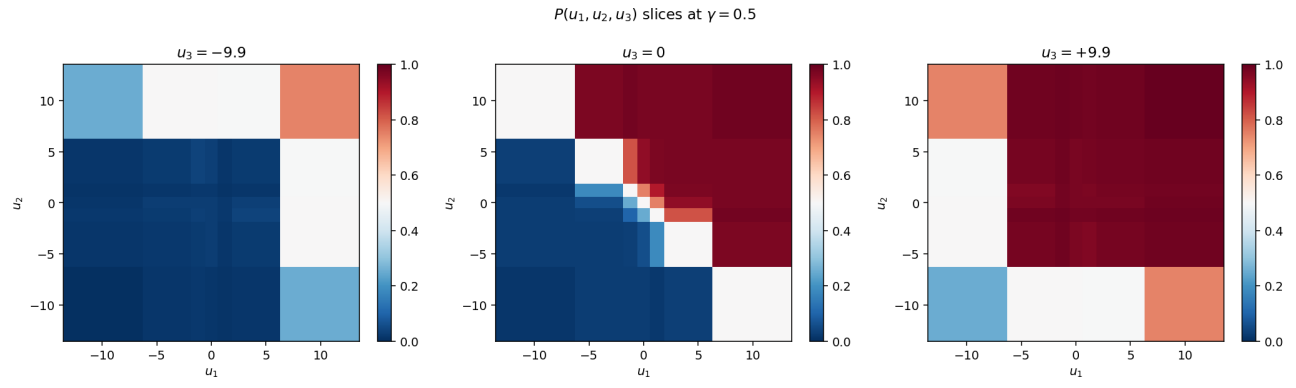
3D price function projected to 2D slices at fixed u_3 .



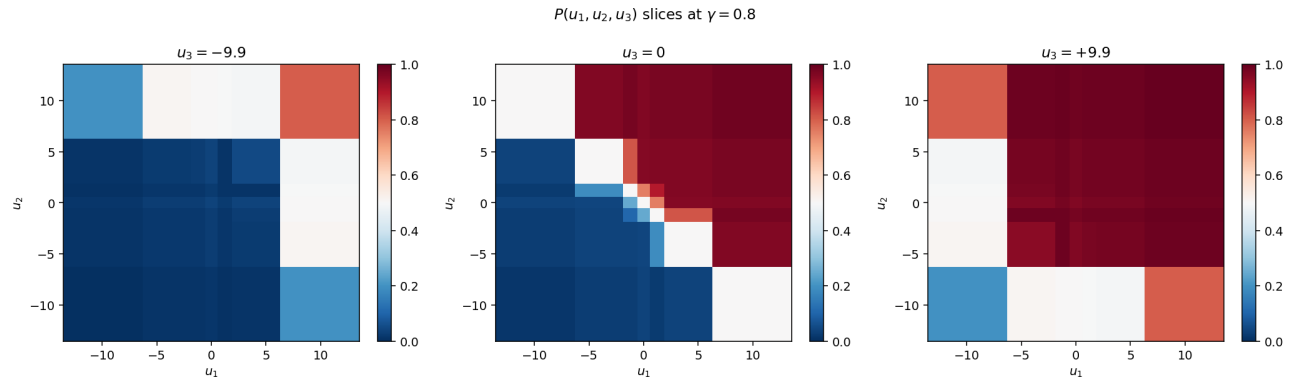
$\gamma = 0.1$: $P(u_1, u_2, u_3)$ at three slices.



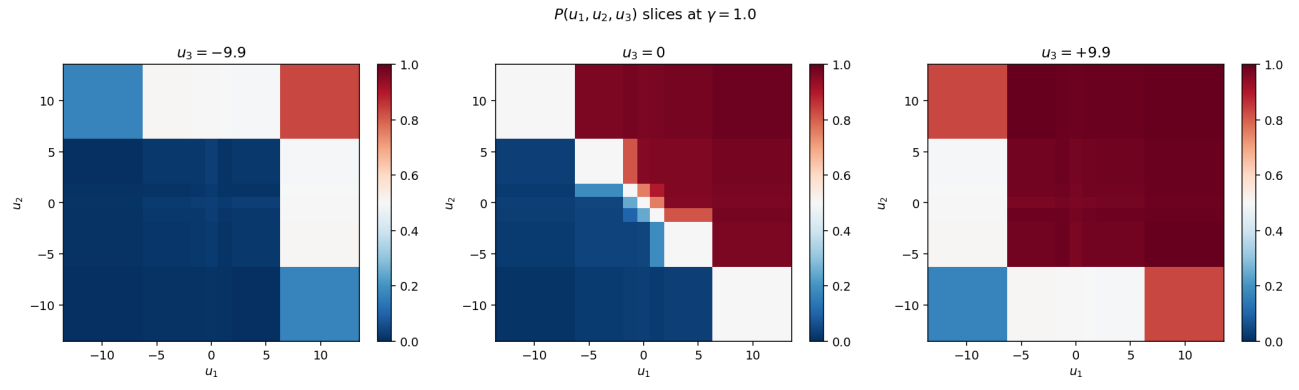
$\gamma = 0.3$: $P(u_1, u_2, u_3)$ at three slices.



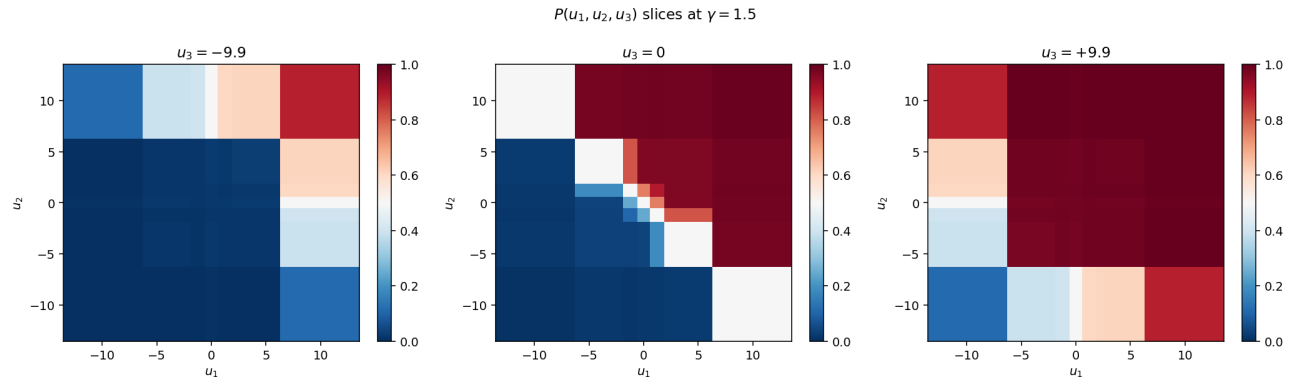
$\gamma = 0.5$: $P(u_1, u_2, u_3)$ at three slices.



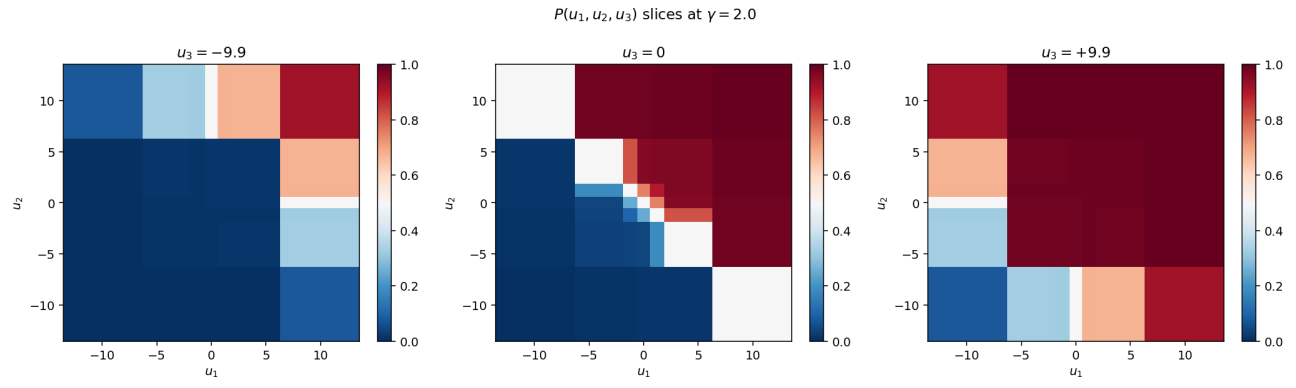
$\gamma = 0.8$: $P(u_1, u_2, u_3)$ at three slices.



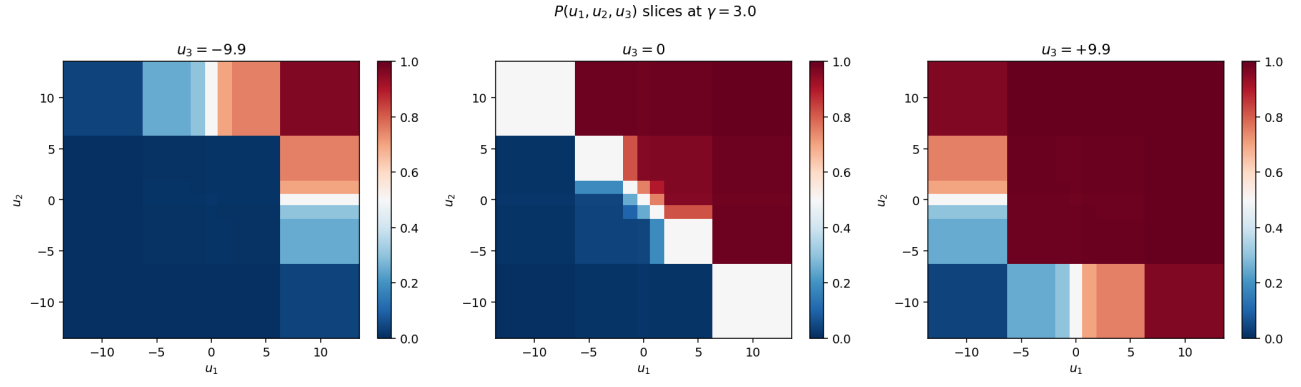
$\gamma = 1.0$: $P(u_1, u_2, u_3)$ at three slices.



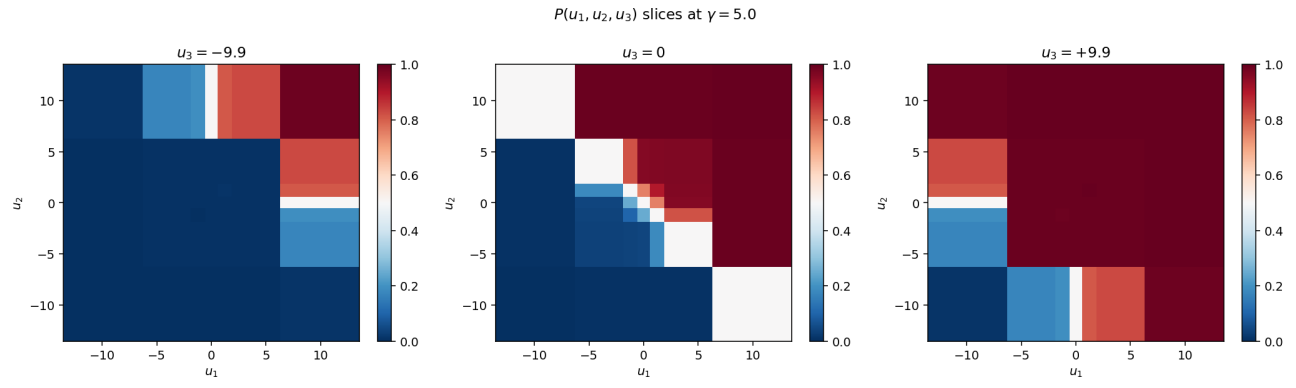
$\gamma = 1.5$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 2.0$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 3.0$: $P(u_1, u_2, u_3)$ at three slices.



$\gamma = 5.0$: $P(u_1, u_2, u_3)$ at three slices.

7 Summary table

γ	best N_Q	$\ F\ _\infty$	slope α^*	deficit $1 - R^2$	solve time
0.1	18	$4.44e - 16$ ★	0.3182	0.1977	410.9 s
0.3	12	$6.43e - 04$	0.3039	0.1668	925.7 s
0.5	16	$1.54e - 03$	0.3179	0.1745	980.6 s
0.8	18	$2.00e - 15$ ★	0.3292	0.1575	634.5 s
1.0	16	$8.88e - 16$ ★	0.3447	0.1448	101.4 s
1.5	12	$1.27e - 03$	0.3675	0.1107	862.5 s
2.0	14	$2.22e - 16$ ★	0.3826	0.1008	167.9 s
3.0	16	$5.76e - 16$ ★	0.4150	0.0895	80.7 s
5.0	12	$9.16e - 16$ ★	0.4493	0.0821	563.0 s

★ = converged to machine ε (6 of 9). The 3 that did not (10^{-3} floor) would need wider N_Q search.